Dongwook **Kwon**

AI Engineer · Agentic AI Systems
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Experience

Suresoft Technologies — *Seongnam, Gyeonggi, Korea*
AI Engineer, Intelligent Agent Technology Team — *Dec. 2025 - Present*
- Developing an AI validation system utilizing AI agent-based architectures.
- Designing and implementing systems to verify and evaluate AI models through autonomous and collaborative AI agents.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University — *Seoul, Korea*
Research Assistant — *Aug. 2021 - Dec. 2025*
- Conducted research on deep learning algorithms, focusing on anomaly detection in time-series data, including Electrocardiogram (ECG) and Cyber-Physical Systems (CPS).
- Participated in the development of an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

LG Electronics Canada — *Toronto, ON, Canada*
Research Project — *Jan. 2025 - Jul. 2025*
- Participating in the development of an assessment framework for agentic AI systems utilizing LLMs to evaluate performance.

Qualcomm Institute, University of California San Diego — *San Diego, CA, USA*
AI Development Project, Summer Program — *Jul. 2022 - Aug. 2022*
- Participated in the Personality-Based Drug Addiction Prediction System project.
- Attended seminars on deep learning and machine learning algorithms.

Education

Kwangwoon University — *Seoul, Korea*
Master of Science in Computer Engineering — *Mar. 2024 - Feb. 2026*
- GPA: 4.33 / 4.5 (98.1%)

University of Toronto — *Toronto, ON, Canada*
Graduate Research Program in Mechanical and Industrial Engineering — *Jan. 2025 - Jul. 2025*
- GPA: 3.68 / 4.0

Kwangwoon University — *Seoul, Korea*
Bachelor of Science in Electronics and Communications Engineering — *Mar. 2019 - Feb. 2024*
- GPA: 3.41 / 4.5 (88.1%)

Honors & Awards

Major Awards

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego — *United States*
Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries — *South Korea*

Academic Recognition

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers — *South Korea*
Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024) — *South Korea*
Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering — *South Korea*

Competitions

Dec. 2025 **Silver Award**, 23rd Embedded Software Competition (KIISE) *South Korea*
Dec. 2022 **Bronze Award**, Hanium ICT Mentoring Competition — FKII *South Korea*
Sep. 2022 **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX) — Kwangwoon University *South Korea*
Oct. 2021 **Grand Prize**, MY (Multi-Y) Capstone Design Competition — Kwangwoon University *South Korea*

Funded Research Projects

LG Electronics Canada *Toronto, ON, Canada*
AGENTIC AI EVALUATION FRAMEWORK *Jan. 2025 - Present*
- Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

Sejong University *Seoul, Korea*
AI-BASED INTRUSION DETECTION FOR APR1400 *Jul. 2022 - Present*
- Developed an AI defense system to protect nuclear power plants from cyber attacks.

Ministry of Science and ICT *Seoul, Korea*
QUADRUPED ROBOT USING VISION SLAM WITH A DEPTH CAMERA *Apr. 2022 - Nov. 2022*
- Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Kookmin University *Seoul, Korea*
ACTIVE KILL-SWITCH AND BIOMARKER-BASED DEFENSE FOR IOT MEDICAL DEVICES *Sep. 2021 - Dec. 2022*
- Developed a security system for heart implant devices using an anomaly detection algorithm.

Ministry of Oceans and Fisheries *Seoul, Korea*
SMART LOGISTICS SYSTEM USING COMPUTER VISION *Apr. 2021 - Nov. 2021*
- Developed a logistics detection system incorporating location tracking and stack-layer identification for containers.

Publications

Journal Articles

Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.

D. Kwon, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery* (Springer Nature), 2025.

Conference Papers

R. Chang†, **D. Kwon**†, J. Lee†, and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026. †Equal contribution.

J. Lee†, R. Chang†, **D. Kwon**†, H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proc. EMNLP 2025 Industry Track (Oral)*, 2025. †Equal contribution.

D. Kwon, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**

D. Kwon, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**

Patents

Issued Patent, "Method for Implantable Medical Device to Detect Anomaly in Real Time," Patent No. KR 10-2961984, Issued on April 30, 2026.

Issued Patent, "GPT API-Based Network Anomaly Detection," Patent No. KR 10-2848814, Issued on August 18, 2025.

Leadership Experience

President, Kwangwoon International Student Association — Lead intercultural programs and student engagement initiatives.

Director of Filming and Editing, Kwangwoon Broadcasting Center (KWBC) — Oversaw media production and content delivery.

Class Representative, Dept. of Electronics and Communications Engineering — Represented student interests in academic affairs.

Executive Member, Amateur Radio Club — Organized radio workshops and technical exhibitions.